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Co-precipitated Fe-Zr catalysts for the Fischer-Tropsch synthesis of lower olefins ($C_2^O \sim C_4^O$): Synergistic effects of Fe and Zr

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ABSTRACT

Synthesis of lower olefins using syngas (H_2/CO) in a fixed-bed reactor under 1.0 MPa, WHSV = 3 $NL_{H_2/CO} \cdot g_{Fe}^{-1} \cdot h^{-1}$ (WHSV = 3 $NL_{H_2/CO} \cdot g_{cat}^{-1} \cdot h^{-1}$ for Zr_p), $H_2/CO = 2$ was investigated over a series of Fe-based catalysts with Zr as the promoter. The catalysts were prepared by co-precipitation (Fe and Zr) and K was impregnated into the precipitate. The prepared catalysts with different Zr/Fe mol ratio (Zr_0, $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p) were characterized by N_2 adsorption/desorption, X-ray diffraction (XRD), Raman spectroscopy (Raman), temperature-programmed reduction (TPR) and X-ray photoelectron spectroscopy (XPS). Synergistic effects of Fe and Zr on the nanostructure, textural, surface property and catalytic performance were analyzed. Characterization of the catalyst showed that the introduction of ZrO_2 is in favor of the formation of mesopores due to the inhibition between Fe and Zr on each other's crystal growth, increasing α-Fe_2O_3 dispersion and the proportion of surface reduction to bulk reduction, which makes the catalysts easily be reduced by H_2 and CO. However, the interaction between Fe and Zr significantly suppresses the formation of θ-Fe_3C and carbon deposition of catalysts. The optimal Fe/Zr mole ratio of Zr_X catalysts is found to be 1/1 due to the synergistic effect of Fe and Zr, the improved distribution of α-Fe_2O_3 and the abundant defects on α-Fe_2O_3 in the catalyst. In the case of Zr_1, the CO conversion and the selectivity toward lower olefins reached 82.1% and 34.6% at 270 °C. Adequate Zr, acting as a block of chain growth and weakening the adsorption of light olefins on active sites, is essential to improve the light olefins selectivity.

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1. Introduction

Ethylene, propylene and butylene (referred to as lower olefins) are the most basic and important organic chemical raw materials in the chemical industry [1,2]. In recent years, synthesis of lower olefins by catalytic conversion of syngas [3–8] has attracted extensive attention. However, there are still some problems and shortcomings needing to be solved in this process, such as interaction between metal components and supports [9], high selectivity of methane [10], high activity of WGSR (Water-gas-shift Reaction) [3,11,12], coke formation on catalyst [13], and so on, which limit the industrialization process of the technology to some extent.

Iron-based catalysts are highly attractive not only because of their relatively low cost but also due to their broad operation conditions and high product selectivity toward lower olefins [14]. Compared with Co catalyst, the ability of chain growth forming long-chain hydrocarbons in the products on Fe catalyst is weak, and thus Fe catalysts are widely used in the study of the reaction of FTO (Fischer-Tropsch to Olefins) [15]. However, crystallite size growth of the iron phase during heat treatment and reaction is one of the main factors for deactivation of commercial catalysts [16].

Alkali, alkaline earth and transition metals are often used as promoters in order to improve the catalytic performance of iron-based catalyst in the FTS (Fischer-Tropsch Synthesis) reaction. K has been widely studied due to its special electron-donating effect and approved to improve the performance of iron-based catalyst in the FTS reaction in a certain content [17].

ZrO_2 is a p-type semiconductor material, which can easily generate oxygen vacancies in the reducing atmosphere and interact with

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active component. Therefore, it can be used as both a support and a co-catalyst. Qiao's group [18] showed that K can act as a promoter when used with iron supported on graphene to enhance the light olefin selectivity. Wang et al. [19] synthesized α-Fe_2O_3 catalysts with high specific surface area and identified that doping Al and Zr could enhance the catalytic performance. Li et al. [20] proposed the addition of Zr improved the turnover frequency and promoted the stability of the catalyst regardless of the incorporation manner. Chen et al. [21] found the addition of small amount of K_2O or ZrO_2 as a promoter significantly changed the FTS performance of 15% Co/MC. However, Li et al. [22] reported the selectivity of $C_2^=$—$C_4^=$ is 27.5% on the $K_{10}Fe_{13}Co_2Zr_{100}$ catalyst and strong synergetic interaction among Fe, Co and Zr provided better metal dispersion and resistance to sintering. Cheng et al. [23] reported hydrogenation of CO were performed over a series of ZrO_2/SAPO bifunctional catalysts through two consecutive processes (methanol synthesis and methanol to olefins) and found that CO was activated in the oxygen vacancies on the zirconia surface and formed surface methoxide via formate in the presence of H_2. However, studies on the effect of ZrO_2 also had conflicting conclusions. For instance, Li et al. [24] reported that Fe would enriched in the surface of Zr-Fe bimetallic catalysts, while Chen et al. [25] found that the surface of Zr-Fe bimetallic catalysts would be enriched with Zr.

Previous studies on Fe-Zr catalysts with low Zr/Fe molar ratio are generally prepared by co-precipitation of iron and zirconia precursors or using ZrO_2 as support to prepared Fe-Zr catalysts with high Zr/Fe molar ratio by impregnation method. Metal dispersion, structural characteristics and the interaction between Zr and Fe have been proved to be the key factors affecting the catalytic performance of Zr-Fe catalysts, due to the synergistic effect between Zr and Fe. The present study aims at studying the synergetic effects between Zr and Fe with different Zr/Fe molar ratios on the co-precipitated Zr-Fe catalysts for the Fischer-Tropsch synthesis to lower olefins. Until now, there have been only a few reports about the synthesis of light olefins by catalytic conversion of syngas over Fe-Zr catalysts. The results indicated that there was a significant synergistic effect on Zr and Fe, the lower olefins selectivity could reach 39.0% when Zr/Fe molar ratio was around 1.

2. Experimental

2.1. Catalyst preparation

The catalysts used in this study were prepared by coprecipitation and impregnation methods. A mixed solution of $ZrOCl_2 \cdot 8H_2O$ and $Fe(NO_3)_3 \cdot 9H_2O$ with a desired Zr/Fe mole ratio (0/1, 0.2/1, 1/1, 2/1, 1/0) were added into aqueous solution of ammonia of 14 wt% at 16 °C drop by drop until pH = 11. The precipitate was washed with deionized water until pH value of filtrate reached 7. Calculated amount of K_2CO_3 was added to 100 mL deionized water to prepare aqueous solution of potassium carbonate. The precipitate was added (Fe/K mol ratio is 100/4 for iron-containing samples, Zr/K mol ratio is 100/4 for non-iron-containing samples) to the above aqueous solution of potassium carbonate with a continuous stirring for 3 h at 16 °C. The resultant slurry was directly dried at 120 °C overnight, powdered and then calcined at 600 °C for 5 h to yield the desired product. Finally, the calcined sample was crushed and sieved with an average size of 20–40 mesh for further tests. In this study, the calcined catalysts are abbreviated as Zr_0, $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p, which represents the atomic ratio of Zr/Fe is 0:1, 0.2:1, 1:1, 2:1, 1:0, respectively.

2.2. Catalyst characterization

The N_2 absorption-desorption, BET surface area, pore volume, and average pore size of fresh catalysts were determined using a Micromeritics ASAP-2020 instrument at 77.35 K. The samples were degassed under a vacuum at 473.15 K for 4 h prior to measurement. Specific areas were calculated by the BET method applied to the region of relative pressure $0.04 < P/P_0 < 0.3$ for adsorption data. The total pore volumes and average pore size were estimated according to the nitrogen adsorption isotherms using the Barrett-Joyner-Halenda (BJH) method.

X-ray powder diffraction (XRD) patterns for the fresh samples were recorded with a Bruker advanced D8 powder diffractometer (Co Kα), and the radiation operating at 40 kV and 40 mA for 2θ angles ranging from 10° to 70°. The crystallite phase was estimated with the data of Joint Committee on Powder Diffraction Standards (JCPDS).

Raman spectra were carried out using a Renishaw RM1000 spectrometer. The exciting laser wavelength was 532 nm under a power of 1.5 mW, and the spectra were recorded in the wave number range of 100–1000 cm^{-1} covering the first-order bands.

Temperature-programmed reduction (TPR) profiles of calcined samples were recorded by a Micromeritics AutoChem2920 instrument using a mixture gas of 7% H_2/93% Ar (H_2-TPR) or 5% CO/95% Ar (CO-TPR), and the consumption of reduction gas was monitored by the change of thermal conductivity of the effluent gas stream using a thermal conductivity detector (TCD).

X-ray photoelectron spectroscopy (XPS) experiments were performed on a VG MiltiLab 2000 spectrometer, and all spectra were obtained using Mg Ka X-ray source. All XPS spectra were obtained from the samples treated in situ in the reaction cell and then transferred into the ultra high vacuum (UHV) chamber without exposure to air. Binding energies (BEs) were calibrated relative to the C 1s peak from carbon contamination of the samples at 284.6 eV to correct for the contact potential differences between the sample and the spectrometer.

2.3. Reactor system and operating procedures

The reactor system was shown in Fig. 1. FTO (Fischer-Tropsch to Olefins) reaction was carried out in a fixed-bed reactor with an internal diameter of 5 mm and a height of 300 mm. In each run, 2.0 g catalyst (particle size: 40–60 mesh) was loaded between two layers of quartz cotton to form a catalyst bed in the reactor. Catalysts were activated using CO (20% CO/ 90% N_2) with WHSV = 5 $NL_{CO} \cdot g_{Fe}^{-1} \cdot h^{-1}$ (WHSV = 5 $NL_{CO} \cdot g_{cat}^{-1} \cdot h^{-1}$ for Zr_p) at 0.5 MPa with temperature increasing from ambient to 300 °C at 5 °C•min^{-1} heating rate, and was maintained at this temperature for 1 h and then reduced to 100 °C. After activation, the system was pressurized to 1.0 MPa by syngas (20% H_2/10% CO/90% N_2, H_2/CO = 2) and maintained at target WHSV (3 $NL_{H_2/CO} \cdot g_{Fe}^{-1} \cdot h^{-1}$ for Zr_0, $Zr_{0.2}$, Zr_1 and Zr_2; WHSV = 3 $NL_{H_2/CO} \cdot g_{cat}^{-1} \cdot h^{-1}$ for Zr_p) for 1 h, and then the temperature was increased to 270 °C at 5 °C•min^{-1}. Hydrogen, carbon monoxide and nitrogen feed gas flow rates were controlled by three separate mass flow controllers. The reaction performance tests were performed at 270 °C, 1.0 MPa and a WHSV of 3 $NL_{H_2/CO} \cdot g_{Fe}^{-1} \cdot h^{-1}$ (WHSV = 3 $NL_{H_2/CO} \cdot g_{cat}^{-1} \cdot h^{-1}$ for Zr_p) for 100 h. The feed and tail gases were analyzed by an on-line gas chromatograph (Agilent 7890B) with a flame ionization detector (FID1) and two thermal conductivity detectors (TCD2 and TCD3). FID1 was used to detect C_1 to C_4 hydrocarbons, TCD2 was used to detect CO, CO_2 and N_2, while TCD3 was used to detect H_2.

The CO conversion (X_{CO}), the selectivity towards CO_2 (S_{CO_2}), hydrocarbons (S_{Cn}, $S_{C_5^+}$, $S_{C_2^oC_4^o}$ and $S_{C_2^pC_4^p}$) were calculated using Eqs. (1)–(6) respectively.

$$X_{CO} = \frac{M_{in}^{CO} - M_{out}^{CO}}{M_{in}^{co}} \times 100\% \tag{1}$$

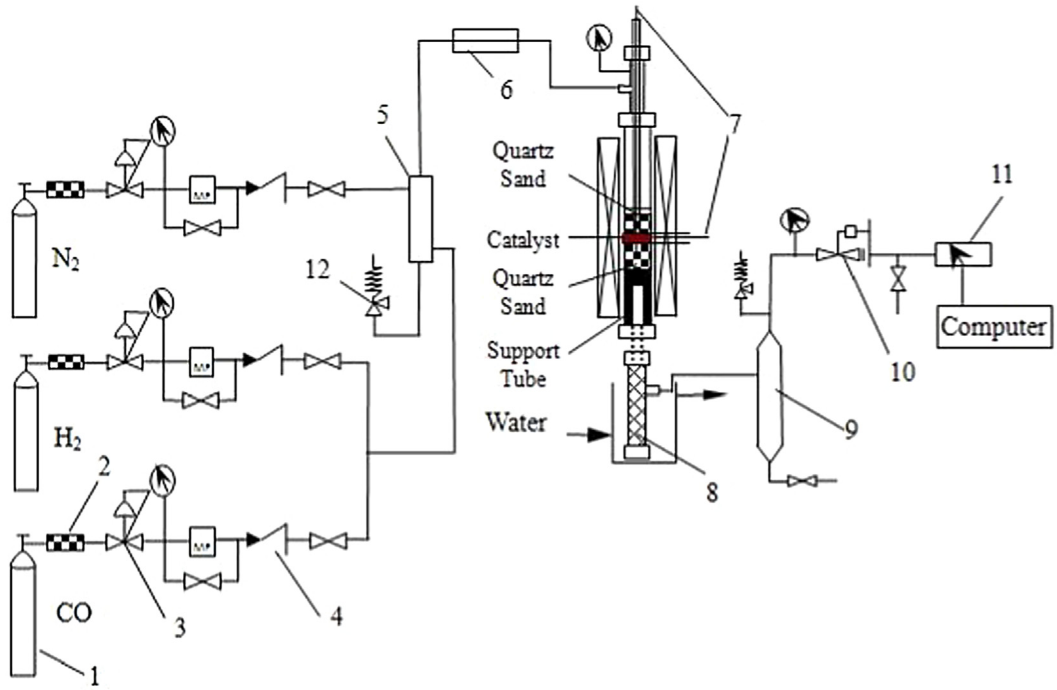

Fig. 1. Schematic diagram of the experimental apparatus for catalyst activity test. 1-steal cylinder; 2-filter; 3-pressure regulator; 4-check valve; 5-gas mixer; 6-preheater; 7-thermocouple; 8-condenser; 9-liquid-gas separator; 10-back pressure valve; 11-gas chromatography; 12-safety valve.

$$S_{CO_2} = \frac{M_{out}^{CO_2}}{M_{in}^{CO} - M_{out}^{CO}} \times 100\% \tag{2}$$

$$S_{Cn} = \frac{n_i M_{out}^{i}}{M_{in}^{CO} - M_{out}^{CO}} \times 100\%, \quad (n = 1, 2, 3, 4) \tag{3}$$

$$S_{C_5^+} = 100 - \sum S_{Cn}, \quad (n = 1, 2, 3, 4) \tag{4}$$

$$S_{C_2^o C_4^o} = \frac{\sum n_i M_{out}^{i}}{M_{in}^{CO} - M_{out}^{CO}} \times 100\% \tag{5}$$

$$S_{C_2^p C_4^p} = \frac{\sum n_i M_{out}^{i}}{M_{in}^{CO} - M_{out}^{CO}} \times 100\% \tag{6}$$

where M_{in}^{CO} is the inlet carbon number of CO, M_{out}^{CO} and $M_{out}^{CO_2}$ are the carbon numbers of CO and CO_2 in tail gas, i is for a particular hydrocarbon, n (n = 1, 2, 3, 4) is the carbon number of hydrocarbon.

3. Results and discussion

3.1. Morphology characterization

Morphology of Zr_x (x = 0, 0.2, 1, 2 and p) is shown by SEM images in Fig. 2. It can be seen that the morphology and particle size distribution of samples changed significantly with the increase of Zr/Fe molar ratio. Fig. 2 (a) and (e) denotes the nanosphere morphology with average particle sizes around 130 nm and 30 nm for Zr_0 and Zr_p respectively. However, the particle size distribution of $Zr_{0.2}$, Zr_1 and Zr_2 shifted toward small particle size (approaching 25 nm) with increasing Zr/Fe molar ratio. Hence, it may be concluded that the addition of ZrO_2 hinders the growth of α-Fe_2O_3 nanoparticles.

The transmission electron microscopy (TEM) image of as-prepared Zr_1 is presented in Fig. 3. The clear fringes of crystal lattice for Zr_1 demonstrate the high crystallinity. The interlayer spacing of 0.27 nm could be assigned to (1 0 4) crystal face of α-Fe_2O_3 and the diffraction spacing at 0.29 nm was attributed to the lattice plane (1 1 1) of ZrO_2. The selected area electron diffraction (SAED) pattern shown in Fig. 3(b) indicated the polycrystalline nature of Zr_1 [26,27].

3.2. Textural analysis

Nitrogen adsorption-desorption isotherms and pore size distribution of the catalysts are shown in Fig. 4. From Fig. 4(a), according to the International Union of Pure and Applied Chemistry (IUPAC) classification, Zr_0, $Zr_{0.2}$ and Zr_1 show type III isotherms, while Zr_2 and Zr_p have type IV isothermals. The type III and IV isothermals are characteristic of mesoporous materials. However, the narrow hysteresis curve (Type H4 loop) presented in the desorption branch of Zr_0 is associated with narrow slit-like pores [28], while the small adsorption capacity combined with the smaller BET surface, pore volume and pore size as shown in Table 1 indicating nearly non-porous nature of Zr_0 [29]. In contrast, the shape of the hysteresis loops for $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p change from Type H4 ($Zr_{0.2}$, Zr_1) to Type H2 (Zr_2, Zr_p), indicating the presence of slit-like pores and ink-bottle pores due to the introduction of ZrO_2. From Fig. 4(b), The pore size distribution for Zr_p cover a range of 7–13 nm, while for Zr_0 only exhibit at 3 nm. By increasing the Zr/Fe mol ratio, the distribution curves shift to smaller pore size and become more concentrated. Meanwhile, the volume of N_2 adsorbed in a high relative pressure range (approaching 1.0) rose rapidly with the increase of the Zr/Fe mol ratio as shown in Fig. 5. These indicate the addition of ZrO_2 is benefit to generate mesopores in spite of possibly derived from the reduction of particle size [30] as shown in Fig. 2.

3.3. XRD of catalysts

The XRD patterns and average crystallite size of the fresh catalysts are presented in Fig. 6(a) and Table 1, respectively. The major

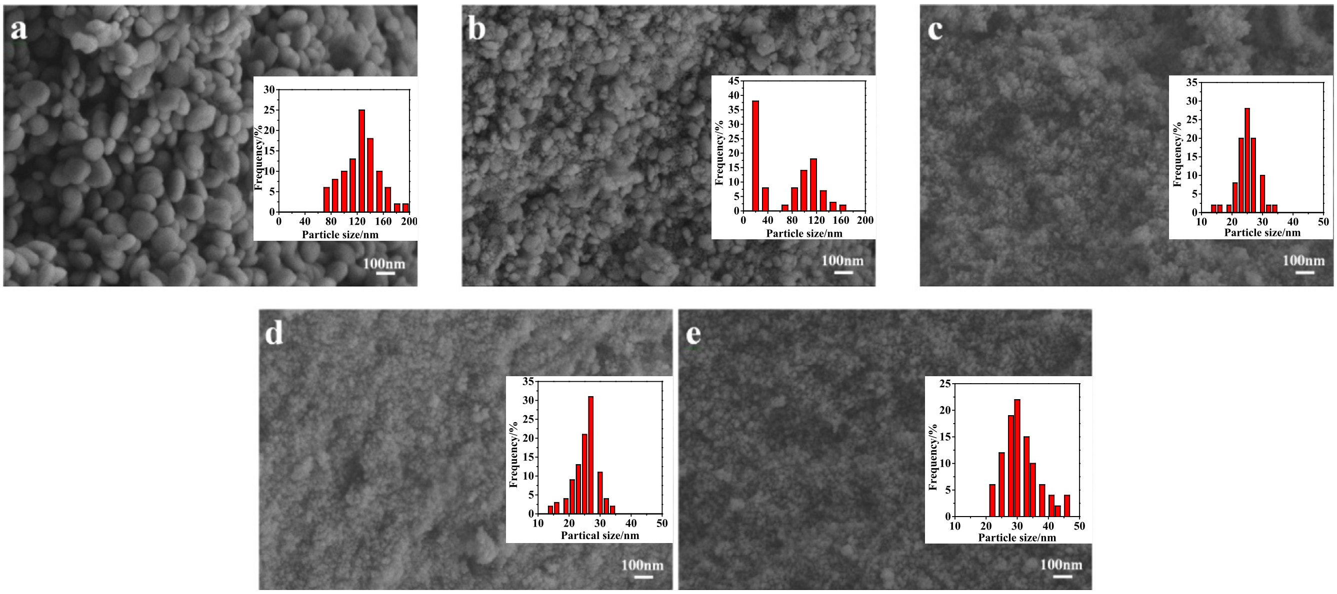

Fig. 2. SEM images and particle size distributions of catalysts: (a) Zr_0; (b) $Zr_{0.2}$; (c) Zr_1; (d) Zr_2; (e) Zr_p.

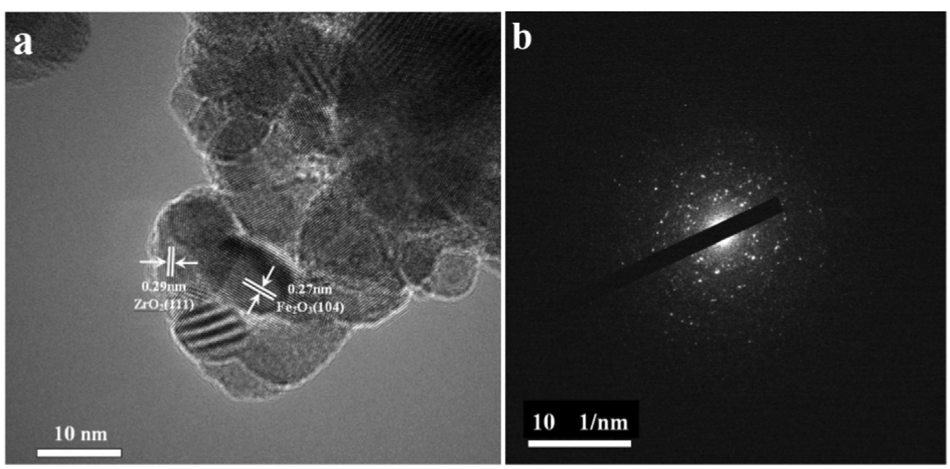

Fig. 3. TEM image (a) and SAED pattern (b) of Zr_1.

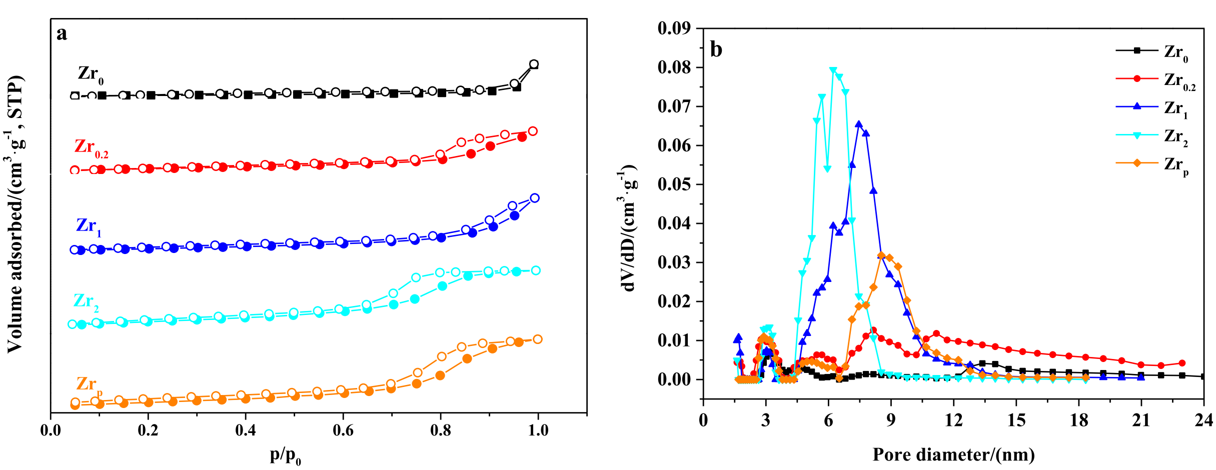

Fig. 4. N_2 adsorption-desorption isotherms of catalysts (a) and corresponding pore size distribution curves (b). Solid and open symbols in the Fig. 4(a) indicates adsorption and desorption points respectively.

Table 1
Textural properties of catalysts and particle size of Fe_2O_3 and ZrO_2 on the catalysts.

Catalyst	BET Surface Area[a] (m^2/g)	Pore Volume[b] (cm^3/g)	Pore Size[c] (nm)	Particle size Fe_2O_3/ZrO_2[d] (nm)	d-spacing ZrO_2 (1 1 1)/α-Fe_2O_3 (1 0 4) (nm)/(nm)
Zr_0	13.59	0.10	3.34	35.9/–	–/0.2690
$Zr_{0.2}$	42.35	0.17	3.33	23.1/4.1	0.2938/0.2702
Zr_1	93.35	0.22	9.68	15.3/9.8	0.2929/0.2697
Zr_2	108.91	0.20	7.82	2.7/10.4	0.2919/0.2695
Zr_p	42.44	0.11	11.98	–/31.3	0.2959/–

[a] : Calculated by BET method.
[b and c] : calculated by BJH method from desorption isothermal.
[d] : calculated by Scherrer equation using (1 0 4) plane and (1 1 0) plane for Fe_2O_3, and (1 1 1) plane and (2 2 0) plane for ZrO_2 from XRD.

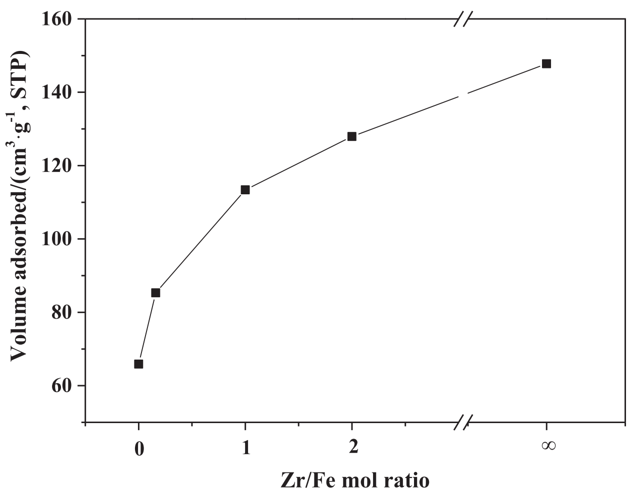

Fig. 5. Effect of Zr/Fe mol ratio on the volume of N_2 absorbed at P/P_0 approaching to 1.0.

diffraction peak of ZrO_2 (JCPDS card no. 89-9096) species belonging to Fm-3 m group were observed, while the rest of the diffraction peaks corresponded to α-Fe_2O_3 (JCPDS card no. 33-0664) with the R-3c space group. As can be observed, all the specimens exhibits a pure crystal of α-Fe_2O_3 and/or ZrO_2, which suggest that K is highly dispersed in the lattice or existed in the amorphous structures [31]. For Zr_0 and Zr_p, only a single phase of α-Fe_2O_3 or ZrO_2 is detected, while the diffraction peak of both α-Fe_2O_3 and ZrO_2 exists in $Zr_{0.2}$, Zr_1 and Zr_2. Moreover, compared with Zr_p, changes of the $Zr_{0.2}$, Zr_1 and Zr_2 2θ value are clearly observed from the characteristic ZrO_2 (1 1 1) diffraction peak as shown in Fig. 6(b). The ZrO_2 (1 1 1) diffraction peak of Zr_p is at 2θ values of 35.16°, corresponding to a d_{111} spacing of 0.2959 nm. However, the d_{111} spacing of ZrO_2 decreased after Fe^{3+} is introduced, and the extent of which is inversely proportional to the Fe/Zr mol ratio, such as 0.2938 nm, 0.2929 nm and 0.2919 nm for $Zr_{0.2}$, Zr_1 and Zr_2 respectively as shown in Table 1 and Fig. 6(b). The results illustrate that Zr^{4+} (r = 0.084 nm) in ZrO_2 was substituted by smaller Fe^{3+} (r = 0.065 nm). Meanwhile, the d_{104} spacing of α-Fe_2O_3 slightly increased from 0.2690 nm to 0.2702 nm, 0.2697 nm and 0.2695 nm for Zr_0, $Zr_{0.2}$, Zr_1 and Zr_2 respectively as shown in Table 1. Fe-Zr mixed oxides ($Fe_xZr_{1-x}O_y$), which prepared by Li et al. [24], showed that the 2θ value of α-Fe_2O_3 (1 0 4) decreased slightly after Zr was doped. Wang et al. [19] studied Fe/Zr catalyst prepared by co-precipitation method, found that α-Fe_2O_3 unit cell would increase slightly, especially in c direction, and ascribed it to the incorporation of Zr^{4+} into α-Fe_2O_3 to form solid solutions. The particle size of Fe_2O_3 decreased gradually with the increase of Zr/Fe molar ratio, but the particle size of ZrO_2 showed an opposite trend, which indicate that Fe and Zr suppressed the formation of each other's grain, which is consistent with SEM images.

3.4. Raman

Fig. 7 displays the Raman spectra of the catalysts. Zr_0 exhibiting six intense vibrational bands (two A_{1g} modes at 224 and 498 cm^{-1}, and four E_g modes at 245, 291, 410 and 610 cm^{-1}, respectively), which are assigned to hematite structure [19]. For tetragonal ZrO_2 six Raman-active modes are expected, namely 147 cm^{-1} (B_{1g}), 261 cm^{-1} (E_g), 318 cm^{-1} (B_{1g}), 463 cm^{-1} (E_g), 611 cm^{-1} (A_{1g}) and 641 cm^{-1} (E_g) [32]. As Zr/Fe mole ratio increases from 0 to 2, the main bands of Fe_2O_3 shift to lower wavenumbers (the red shift, hence lower energy) for Zr_0, $Zr_{0.2}$, Zr_1 as shown in Fig. 6 (b). However, the bands of Fe_2O_3 shift from 220.7 cm^{-1} to

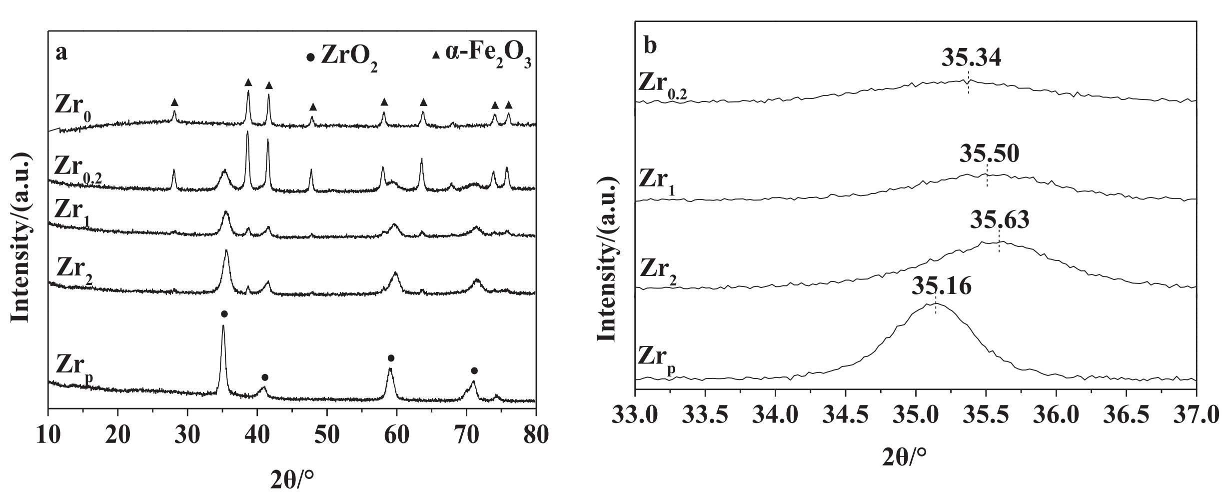

Fig. 6. XRD pattern (a) of catalysts and amplification pattern (b) in the range of 33–37°.

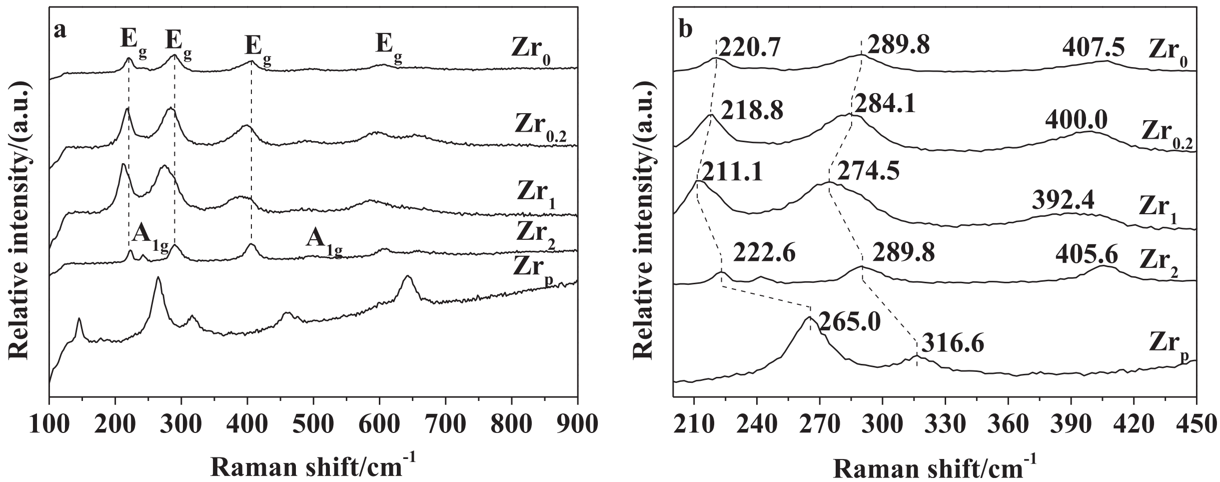

Fig. 7. Raman patterns (a) of catalysts and amplification pattern (b) in the range of 200–440 cm^{-1}.

222.6 cm^{-1} for Zr_2. The blue shift in Raman peaks toward higher wavenumbers can be attributed to the decrease in interionic distance between Fe and O [33], while the red shift toward lower wavenumbers due to changes in particle size, surface strain and formation of defects [34]. In this study, the incorporation of Zr^{4+} into the Fe_2O_3 lattice could induce the defects, which would influence the crystal symmetry and make the Raman active mode of the Fe_2O_3 structure in catalysts shift from that in the pure Fe_2O_3 [35]. Appropriate amount of ZrO_2 is benefits the formation of defect, which facilitate electron transfer from ZrO_2 to Fe_2O_3 since Fe (1.83) is more electronegativity than Zr (1.33). The shifting of Raman bands also indicates the change of surface properties that can be referred to the following XPS analysis.

3.5. X-ray photoelectron spectroscopy (XPS)

The XPS spectra of Fe 2p are shown in Fig. 8. Fe 2p XPS spectra indicate changes in oxidation state. The binding energy (BE) variation observed for Fe 2p indicates the substitution of Zr at Fe site. The Fe (2p) peaks for $Zr_{0.2}$ sample shifted to lower BE at 709.8 eV compared with that of Zr_0 at 710.2 eV. Suo et al. [36] pointed out that if bonding to silica transferred an electron to the Fe species, the Fe BE would shift to lower energy, and if bonding to silica transferred an electron from the Fe species, the Fe BE would shift to higher energy. It is noted that replacing ferric iron with zirconium makes nearby Fe^{3+} turned into Fe^{2+} [37], which may act as electron transfer due to charge compensation effect. However, the Fe (2p) peaks for Zr_1 and Zr_2 shifted to higher BEs compared with $Zr_{0.2}$, which can be ascribed to the reduced coordination number of surface Fe atoms due to smaller size of nanoparticles [37]. It must be mentioned that the satellite peak of Fe $2p_{3/2}$ for Fe_2O_3 located around 718 eV gradually weakens with the increase of Zr/Fe molar ratio. Yamashita et al. [38] observed that the BEs of Fe $2p_{3/2}$ and Fe $2p_{1/2}$ are 711.0 eV and 724.6 eV with a satellite peak at 718.8 for Fe^{3+}; the BEs of Fe $2p_{3/2}$ and Fe $2p_{1/2}$ are 709.0 eV and 722.6 eV with a "shoulder" peak at 714.7 for Fe^{2+}; the BEs of Fe $2p_{3/2}$ and Fe $2p_{1/2}$ are 710.6 eV and 724.1 eV with no satellite for Fe_3O_4. It suggests that Fe^{3+} in the lattice of Fe_2O_3 is substituted by Zr^{4+}, and Fe_2O_3 is converted to Fe_3O_4 on the surface of samples.

The Zr 3d XPS spectra of $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p are shown in Fig. 9. The peak of Zr $3d_{5/2}$ is located at 181.6 eV, corresponding to Zr^{4+} ions in substitutional mode [39], due to it locates between that of ZrO_2 (182.2 eV) [15] and metallic Zr (178.7 eV). All the catalysts displayed that the BEs shift to lower values with increasing Fe content only except for Zr_2, which shows almost identical to Zr_p. Theoretically, Zr^{4+} in the lattice of ZrO_2 is substituted by Fe^{3+}, Zr atoms can release electrons from the outer layer to Fe atoms due to smaller electronegativity of Zr than Fe, resulting in an decrease of electron cloud density in the outer layer of Zr. Hence, the shielding effect of outer electrons on inner electrons decreases, and the BE of outer layer on the inner electron increases, leading to a shift of Zr $3d_{5/2}$ and Zr $3d_{3/2}$ from 181.6 eV and 183.9 eV to higher values

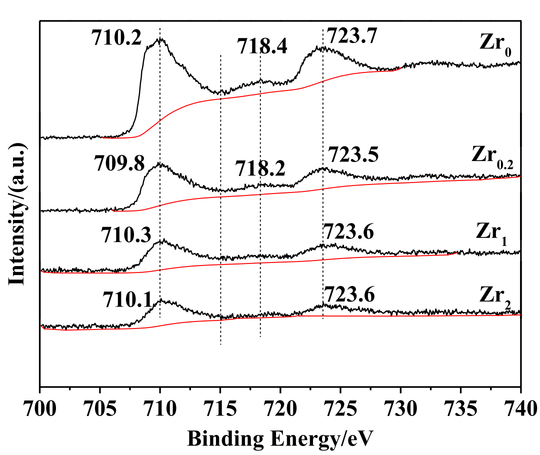

Fig. 8. Fe 2p XPS of catalysts.

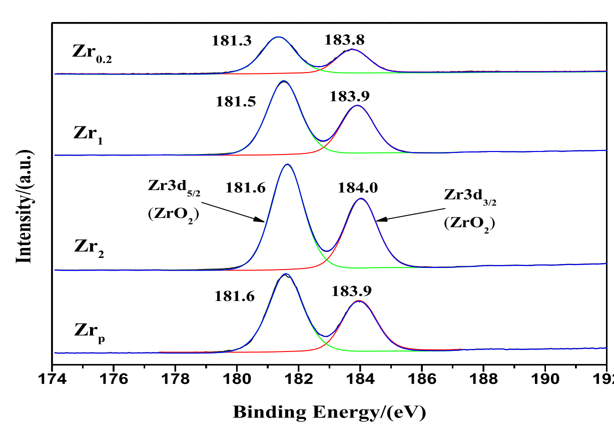

Fig. 9. Zr 3d XPS of catalysts.

[40]. However, through the reports of Sinhamahapatra et al. [41], Wang et al. [42] and Gu et al. [39], it can be concluded that with the increase of oxygen vacancy in ZrO_2 nanoparticles, the peaks of Zr $3d_{5/2}$ and Zr $3d_{3/2}$ will negative broaden and shift to lower BEs. Based on these studies, it can be speculated that with the decrease of Zr/Fe molar ratio, more surface defects via oxygen vacancies will be generated on ZrO_2 nanoparticles, resulting in the decrease of BEs, although the determination of BE shifts and its relationship with bond identities, dopants, defects and other factors are not very clearly [43].

The surface composition of catalysts calculated from XPS and ICP data are shown in Table 2. The bulk Zr/Fe atomic ratio is significantly higher than that on the surface, indicating that Zr is enriched on the catalyst surface, which is consistent with Chen's results [25].

3.6. TPR of catalysts

Fig. 10 shows the H_2-TPR profiles of catalysts. All samples mainly consisted of three peaks: a, b and c. For Zr_0, $Zr_{0.2}$, Zr_1 and Zr_2, the first hydrogen consumption peak (a) is lower than 500 °C representing the initial reduction of α-Fe_2O_3 to Fe_3O_4 and the second peak (b) at around 600 °C is assigned to the reduction of Fe_3O_4 to FeO [44]. However, the third one (c) showed a broad reduction feature at temperatures around 750 °C including the following stages simultaneously: (i) (FeO) → α-Fe and (ii) FeO → Fe_3O_4→ (FeO) → α-Fe, because the FeO is unstable and easily decomposes into metallic Fe and magnetite without hydrogen consumption [25]. The TPR profile of pure ZrO_2 was flat due to the high stability of the oxide below 1000 °C. Peak deconvolution was applied to fit the experimental data and 2–3 sections could be well fitted in each peak as shown in Fig. 10. The section (a_1) might be the reduction of small α-Fe_2O_3 particles to Fe_3O_4 with high dispersion on the surface of the catalysts, while the section (a_2 and a_3) was for the reduction of bulk α-Fe_2O_3 to Fe_3O_4 [25]. In this study, the a_1/a_2 area ratios is applied to qualitatively described the ratios of surface phase to bulk phase in Zr_0, $Zr_{0.2}$, Zr_1 and Zr_2, and results are illustrated in Table 3. It is clearly that the a_1/a_2 values increased with the increasing of Zr/Fe mole ratio, evidencing that the proportion of surface α-Fe_2O_3 reduction to Fe_3O_4 increases with increasing the Zr/Fe mole ratio in samples. Combined with Fe_2O_3 particle sizes in Table 1, it is clear that with increasing Zr/Fe mole ratio, the α-Fe_2O_3 particle size decreased, which resulted in the increase proportion of surface reduction to bulk reduction since the increase of surface α-Fe_2O_3 and the decrease of bulk α-Fe_2O_3.

With the addition of zirconia, TPR profiles become complicated. The reduction peaks of a_1 and a_2 first shift to higher temperatures for $Zr_{0.2}$, and then shift to lower temperatures for Zr_1 and Zr_2 compared with Zr_0. Similar results also has been discovered by Chen et al. [45] over the iron-zirconium oxides prepared by the coprecipitation method with different iron content. This result indicates that the zirconia addition has complex effects on the reduction behavior of iron oxides in catalysts. First, a small amount of zirconia addition inhibits the reduction of iron oxides. This inhibitive effect is due to the strong interactions between Fe^{3+} and zirconia [45]. Chen et al. [25] have reported that the Fe^{3+} cations doped in the zirconia lattice facilitated the formation of Fe-Zr mixed oxide, and then strengthened the interaction between iron and zirconia, leading to the decrease of reducibility of the catalysts. Second, a large amount of zirconia improves the reduction of partial Fe species. As displayed in Table 2, the particle size of Fe_2O_3 is decreased by zirconia. According to the work of Wyrwalski et al. [46], reduction of the small Fe_2O_3 particles is easier than that of the bulk Fe_2O_3 phase. Therefore, the decreased reduction temperature for Zr_1 and Zr_2 might be attributable to the somewhat smaller α-Fe_2O_3 particle size.

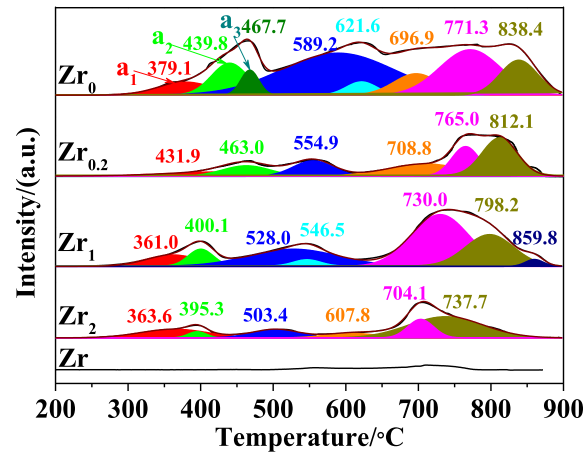

Fig. 10. H_2-TPR curves of the catalysts.

Lots of reports [10,11,14,15] have illustrated that in the process of activation and reaction, the bulk α-Fe_2O_3 is reduced to Fe_3O_4 and the surface α-Fe_2O_3 is transformed to Fe_xC, which formed the state of catalyst and confirmed as its "work type". It is widely accepted that the metallic iron or iron carbides are active for FTS reaction while Fe_3O_4 is active for water-gas shift reaction (WGS). Transformation behavior of iron structures on the surface layers is further investigated by CO-TPR as shown in Fig. 11. The initial reduction temperature of each sample decreased in CO-TPR compared with that in H_2-TPR, which is due to the stronger reduction ability of CO than H_2 [47]. It was found that there are three steps in the CO-TPR reaction, the first peak below 350 °C corresponded to the reduction processes of α-Fe_2O_3 to Fe_3O_4 and carbide formation stages of ε-carbides (ε-Fe_2C and ε′-$Fe_{2.2}C$, below 250 °C) and χ-Fe_5C_2 (between 250 and 350 °C), the second peak region between 350 and 700 °C is a complex reactions corresponds to the reduction of Fe_3O_4 to α-Fe, carbide formation stage of θ-Fe_3C and Boudouard reaction (CO → C + CO_2), etc [48,49], and the third peak above 700 °C corresponded to the carburization of the difficultly reduced iron oxide phase [50]. High CO conversion rate and hydrocarbon

Table 2
XPS results of catalysts.

Catalysts	Surface atom content[a] (atomic %)				Zr/Fe atomic ratio	
	Zr	Fe	K	O	Surface[a]	Bulk[b]
Zr_0	0	21.92	4.56	73.52	0	0
$Zr_{0.2}$	11.74	10.38	2.74	75.14	1.13	0.24
Zr_1	19.87	5.89	1.85	72.39	3.37	1.10
Zr_2	21.90	5.74	0.57	71.79	3.82	2.16
Zr_p	24.26	0	5.22	70.52	–	–

[a] Calculated from XPS.
[b] Calculated from ICP.

Table 3
a_1/a_2 values of Zr_0, $Zr_{0.2}$, Zr_1 and $Zr_{2.}$

Catalysts	Zr_0	$Zr_{0.2}$	Zr_1	Zr_2
a_1	65.98	44.07	55.34	54.77
a_2	149.91	46.98	39.82	10.71
a_1/a_2	0.44	0.94	1.39	5.11

Calculated by the peak areas obtained from H_2-TPR.

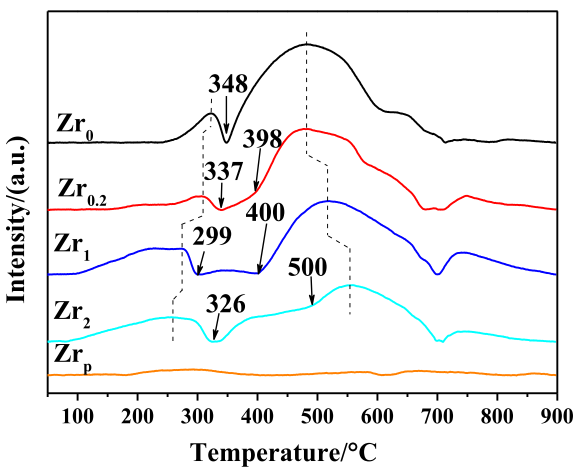

Fig. 11. The CO-TPR curves of the catalysts.

productivity have been associated with χ-Fe_5C_2 while catalyst deactivation due to carbon deposition can be associated with θ-Fe_3C [51]. The first peak gradually move to lower temperature with the increasing of Zr/Fe mole ratio, which also caused by the fact that the Fe_2O_3 particle sizes are decreased with the increase of Zr/Fe mole ratio, since iron oxide with little particle size can be easily reduced. However, it is interesting that the second peak gradually move to higher temperature except for $Zr_{0.2}$, which maintained nearly the same compared with Zr_0. It is reported [52] that metals have different effects on the carburization of iron-based catalysts and this probably due to the difference of metals in enhancing CO dissociation adsorption of catalysts. In this study, it can be inferred that the addition of zirconia can effectively inhibit the reduction and carbonization of catalysts, thus inhibiting the formation of θ-Fe_3C and carbon deposition through transformation from ε-carbides and/or χ-Fe_5C_2 to θ-Fe_3C and Boudouard reaction, and which is consistent with the results of Zhang et al. [50].

3.7. Catalytic performance

The catalytic performance of the catalysts during the FTO reaction was evaluated at 270 °C and 1.0 MPa to test the effects of the Zr promoter. The results of CO hydrogenation are summarized in Table 4. It can be seen that the CO conversion, CO_2 and hydrocarbon selectivity are affected by Zr content. The CO conversion rose from 70.2% (Zr_0) to 90.7% ($Zr_{0.2}$), and then decreased rapidly to 82.1 (Zr_1) and 53.2% (Zr_2). This result suggests that small amount of ZrO_2 is conductive to the increase of BET surface area and pore volume, and the reduction of α-Fe_2O_3 particle size, which increase the contact opportunity between the active site and the reactants. However, as listed in Table 2, excess surface zirconium species might hinder the catalytic activity for CO hydrogenation. The low CO conversion (1.2%) of Zr_p indicates that zirconia can be used not only as support but also as co-catalyst due to the adsorption and activation of CO on the oxygen defect sites [22,40,53,54] to form surface methoxide in the presence of H_2 [23].

According to the selectivity calculated by carbon fraction, the selectivity towards CO_2, total $C_2^O \sim C_4^O$, total $C_2^P \sim C_4^P$ and O:P peaked at Zr_1 (48.2%, 34.6% , 4.8% and 7.2, respectively), while the selectivity toward C_5^+ reached the lowest point (3.0%), with increasing Zr/Fe molar ratio. The selectivity towards total $C_2^O \sim C_4^O$ and total $C_2^P \sim C_4^P$ shows reverse trend with C_5^+, which is consistent with the findings of Chen et al. [25]. Ahmad et al. [29] studied the Zr-pillared montmorillonite supported cobalt nanoparticles catalysts, and found that the acidic nature of zirconia play an important role in hydrocraking reaction, which convert higher molecular weight hydrocarbons to lower hydrocarbons. In this case, the surface acidity of zirconia promotes the hydrocracking of C_5^+ or higher Zr content may cut off the neighboring iron active sites and block C-C coupling to long chain hydrocarbons, which was beneficial to the formation of small molecular hydrocarbons. In addition, González-Castaño et al [55] reported that Zr could easiness of carbonaceous species desorption in Pt/CeZrFeAl catalysts during WGS. Therefore, it can be speculated that with the increase of Zr content, the adsorption capacity of the active site surface to hydrocarbons (whether generated by CO hydrogenation or higher molecular weight hydrocarbons decomposition) decreases, which reduces the probability of the secondary olefin hydrogenation and eventually leads to the increase of O:P. The selectivity toward C_5^+ reached 95.4% for Zr_p. According to previous research [23,40,55–57], it can be found that ZrO_2 can catalyze the conversion of syngas into oxygen-containing compounds (such as methanol, HCOO species, dimethyl ether and so on), which was trapped in the liquid-gas separator and cannot be detected. So that, the results of C_5^+, which was calculated by the formula of Eq. (4), include all undetected

Table 4
Catalytic performance of Zr_0, $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p under conditions of 270 °C, 1.0 MPa, H_2/CO = 2, CO conversion, CO_2 selectivity, hydrocarbon selectivity obtained at 10 h time on stream.

Catalyst	Zr_0	$Zr_{0.2}$	Zr_1	Zr_2	Zr_p
Total CO conversion/%	70.2	90.7	82.1	63.2	1.2
C selectivity into CO_2/(C %)	5.3	29.8	48.2	19.3	1.0
Hydrocarbon selectivity/(C %)					
C_1	1.3	14.6	9.4	4.5	3.4
C_2^O	0.5	4.8	17.9	2.3	0.2
C_2^P	0.1	1.0	3.4	0.5	0
C_3^O	0.4	5.1	15.4	1.2	0
C_3^P	0.1	0.6	1.4	0.4	0
C_4^O	0.2	0.3	1.3	0	0
C_4^P	0	0.1	0	0	0
C_5^+	92.1	43.8	3.0	71.8	95.4
Total $C_2^O \sim C_4^O$	1.0	10.2	34.6	3.5	0.2
Total $C_1^P \sim C_4^P$	1.5	16.2	14.2	5.4	3.4
O:P C ratio in $C_2 \sim C_4$	5.5	6.3	7.2	3.9	∞

Note: ^{O} and ^{P} represent alkenes and alkanes, respectively.

substances. So, adequate Zr that acts as a block of chain growth and weakens adsorbs between active sites and light olefins are essential to improve light olefins selectivity.

The CO_2 selectivity is firstly increased from 5.3% (Zr_0) to 48.2% (Zr_1), then drops rapidly over Zr_2 (19.3%) and Zr_p (1.0%). It has been reported that during the FTS process, the bulk Fe_3O_4 and the surface Fe_xC coexist in the catalyst, and the metallic iron or iron carbides are active for FTS reaction while Fe_3O_4 is active for WGSR. The surface α-Fe_2O_3 is apt to be carbonized to iron carbides, while the bulk α-Fe_2O_3 is tend to be reduced to Fe_3O_4 during activation process and FTS. With increasing of Zr/Fe molar ratio, the surface α-Fe_2O_3 increased while the bulk decreased due to the reduction of the particle size of Fe_2O_3 as shown in Table 1. In this study, a proper amount of ZrO_2 is beneficial for the reaction of WGSR, and which will increase the H_2/CO ratio and CO_2 concentration in the reactor, while too much ZrO_2 is not benefit the generation of CO_2, and which may be caused by the decrease of bulk Fe_3O_4. Li et al. [58] found that a high H_2/CO ratio can facilitate the production of methane, but has a negligible influence on the product distribution. Therefore, the increase of CH_4 selectivity with the increasing of Zr content in the range of Zr/Fe = 0.20 and Zr/Fe = 1.0 may also be affected by WGSR, which produce more H_2 in the reactor than in the feed gas.

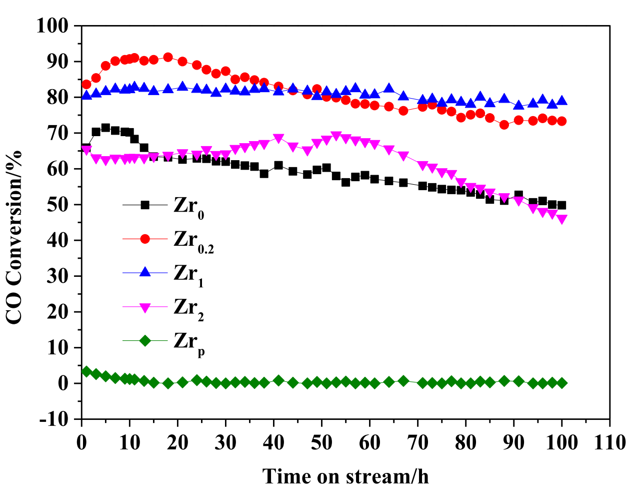

Fig. 12. CO conversion with time on stream over Zr_0, $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p. Reaction conditions: 270 °C, 1.0 MPa, H_2/CO = 2, WHSV = 3 $NL_{H_2/CO} \cdot g_{Fe}^{-1} \cdot h^{-1}$ (WHSV = 3 $NL_{H_2/CO} \cdot g_{cat}^{-1} \cdot h^{-1}$ for Zr_p).

The CO conversion as a function of reaction time for Zr_0, $Zr_{0.2}$, Zr_1, Zr_2 and Zr_p at 270 °C for 100 h is shown in Fig. 12. The CO conversion of Zr_0, $Zr_{0.2}$, Zr_1 and Zr_2 was initially low and gradually increased before reaching the peak at around 5 h, 10 h, 30 h and 50 h respectively. Afterward, Zr_0, $Zr_{0.2}$, Zr_1 and Zr_2 showed different stability pattern within the rest time on stream. Connell et al. [59] and Niu et al. [60] reported that the potassium species migrate to the surface of the catalysts due to a lack of enough oxygen sites in the bulk catalyst to accommodate the larger potassium ions under reaction condition. Niu et al. [61] also pointed out that a low amount of K is enough to promote the FTS activity and stability, while excessive K-loading tends to notably enhance the deactivation rate. Therefore, the unstable of the catalytic performance probably resulting from surface enrichment of potassium on the catalysts surface. The CO conversion of Zr_p decreased rapidly from 3.3% at 1 h to 1.0% at 11 h, and then stabilized below 1.0%. As shown in Fig. 12, $Zr_{0.2}$ and Zr_1 exhibited high activity and stability compared with Zr_0 and Zr_2. This result indicates that the addition of ZrO_2 could improve the stability of catalysts, which probably due to the acidic nature of zirconia promote the conversion of higher molecular weight hydrocarbons to lower hydrocarbons [29], and/or inhibit the transformation from ε-carbides and/or χ-Fe_5C_2 to θ-Fe_3C and Boudouard reaction as shown in TPR, resulting in the loss of carbon deposition on the surface of catalysts. However, Zr_2 suffered a rapid deactivation after 50 h. Carbon deposition and transformation of iron carbides (ε-carbides → χ-Fe_5C_2 → θ-Fe_3C) are considered as a potential deactivation mechanism, which may occurring in parallel [51]. Pour et al. [62] pointed out that the most catalytic active phase in FTS may be ε-carbides, which will last over long periods of reaction, convert into the active χ-Fe_5C_2, and subsequently convert into the inactive θ-Fe_3C. During the phase transformation, carbon atoms may precipitated on the surface and serve as the nucleation sites for polymerization to C_β and/or C_δ. Thus, there is possibility of pore blockage due to the inactive C_δ. Therefore, it can be speculated that some of exposed active sites may covered by carbon deposition, while the covered active sites is no longer assessable for syngas, resulting in a gradual decline in CO conversion.

Based on the above analysis, the speculated catalytic mechanism of the catalysts and the synergistic effect of Fe and Zr are shown in Fig. 13. The incorporation of Zr could observably increase the BET surface area, pore volume, pore size and decrease the par-

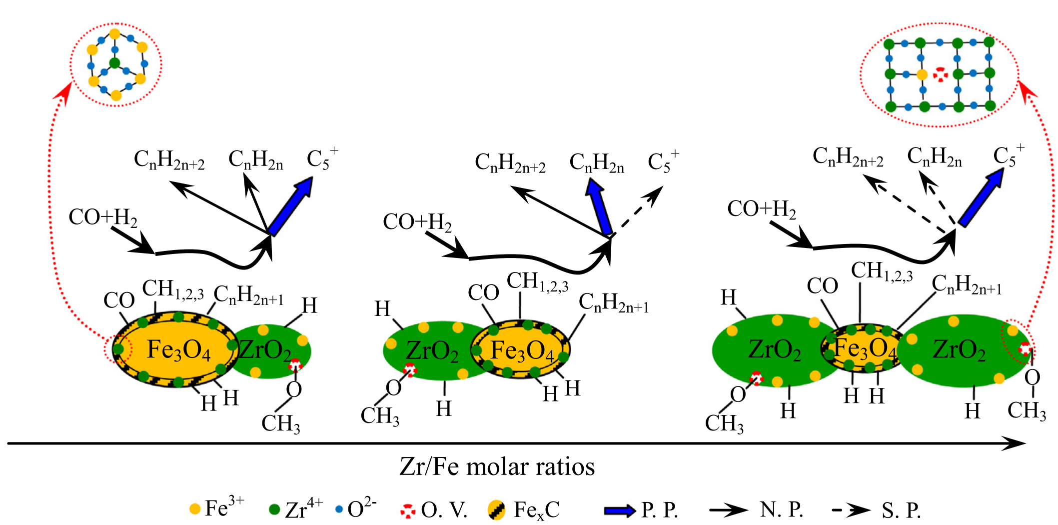

Fig. 13. Schematic illustrations of the speculated catalytic mechanism and the synergistic effect of Fe and Zr.

ticles size of α-Fe_2O_3, which is benefit to improve catalytic activities through increasing FTO reaction area sites. However, excessive ZrO_2 will cause the surface of Fe_2O_3 particles to be covered by ZrO_2, and more surface Fe^{3+} will be replaced by Zr^{4+}, resulting in a decrease in the number of surface active sites and the CO conversion rate during FTO reaction. With the increase of Zr/Fe molar ratio, the adsorption capacity of the surface active sites on the carbonaceous compounds (including CO) gradually decreases, leading to the decrease of hydrogenation reaction capacity, and which will influence the predominant product (P. P.), normal product (N. P.) and slight product (S. P.). As a result, the predominant product of the reaction gradually changes from C_5^+ to $C_2^O \sim C_4^O$, and then to C_5^+.

4. Conclusions

In summary, catalysts with different Zr/Fe molar ratio were tested and compared under the same conditions. The molar ratio of Zr/Fe equals to 1 offered the highest lower olefins selectivity. The increased selectivity should be attributed to the synergistic effect of Zr and the strong interactions among Zr and Fe. In particular, XRD, Raman, the N_2 absorption-desorption, H_2-TPR and CO-TPR indicated that the catalytic performance was attributed to structure properties, crystal configuration and interactions between iron oxides and zirconia. SEM revealed that the incorporation of Zr could significantly reduce the particle size. Raman, TEM and XPS verified that the surface chemical composition greatly changed due to the introduction of Zr. In addition, appropriate amount of zirconia can significantly promote the formation of light olefins, inhibit the formation of C_5^+, and improve the stability of catalysts.

Acknowledgements

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